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Roll No

ME-701(C)-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Industrial Robotics

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Define a Robot? Discuss its need and importance?
b) Enlist the applications and characteristics of Industrial Robots?
2. If two links of a 2-link planar manipulator have equal lengths, find out the expression for the homogeneous transformation matrix.
3. Explain the principle of operation of pneumatic grippers and suggest the types of cargo for which they would be most suited.
4. What is Jacobean of a robot system? Derive the Jacobean matrix for the question stated above.
5. What are the characteristics one should check while selecting a sensor?

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6. What is the criteria for selection of a SENSOR for a robot.
Describe in brief a Piezoelectric sensor.
7.
 - a) What is quantization error? How it can be corrected or minimized?
 - b) Make a list of limitations of two-dimensional vision systems.
8. Describe the terms :
 - a) Range sensing
 - b) Image processing
 - c) Object recognition
